

Random Walks

Course: Practical Time Series Analysis

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March 24, 2024

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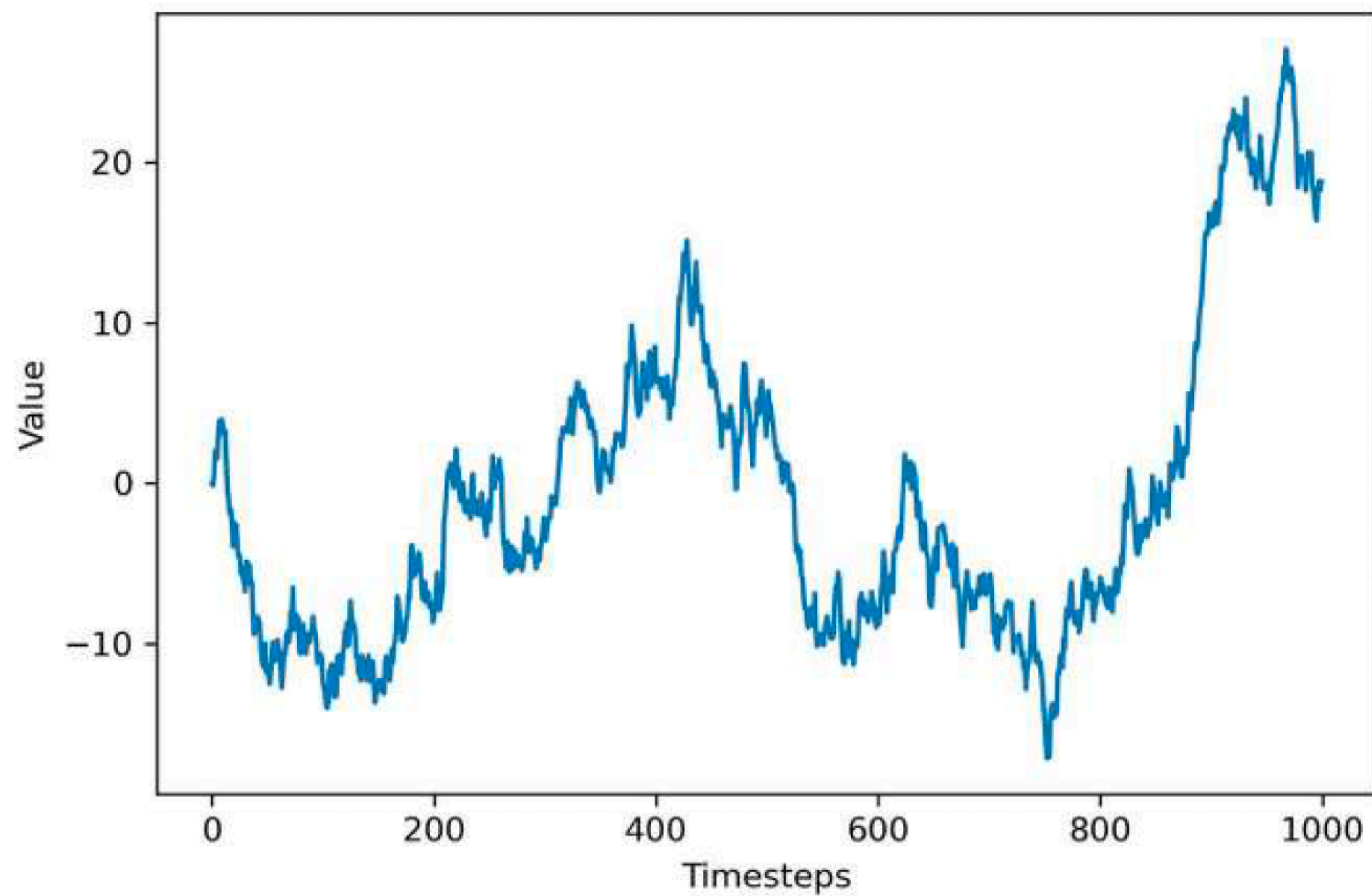
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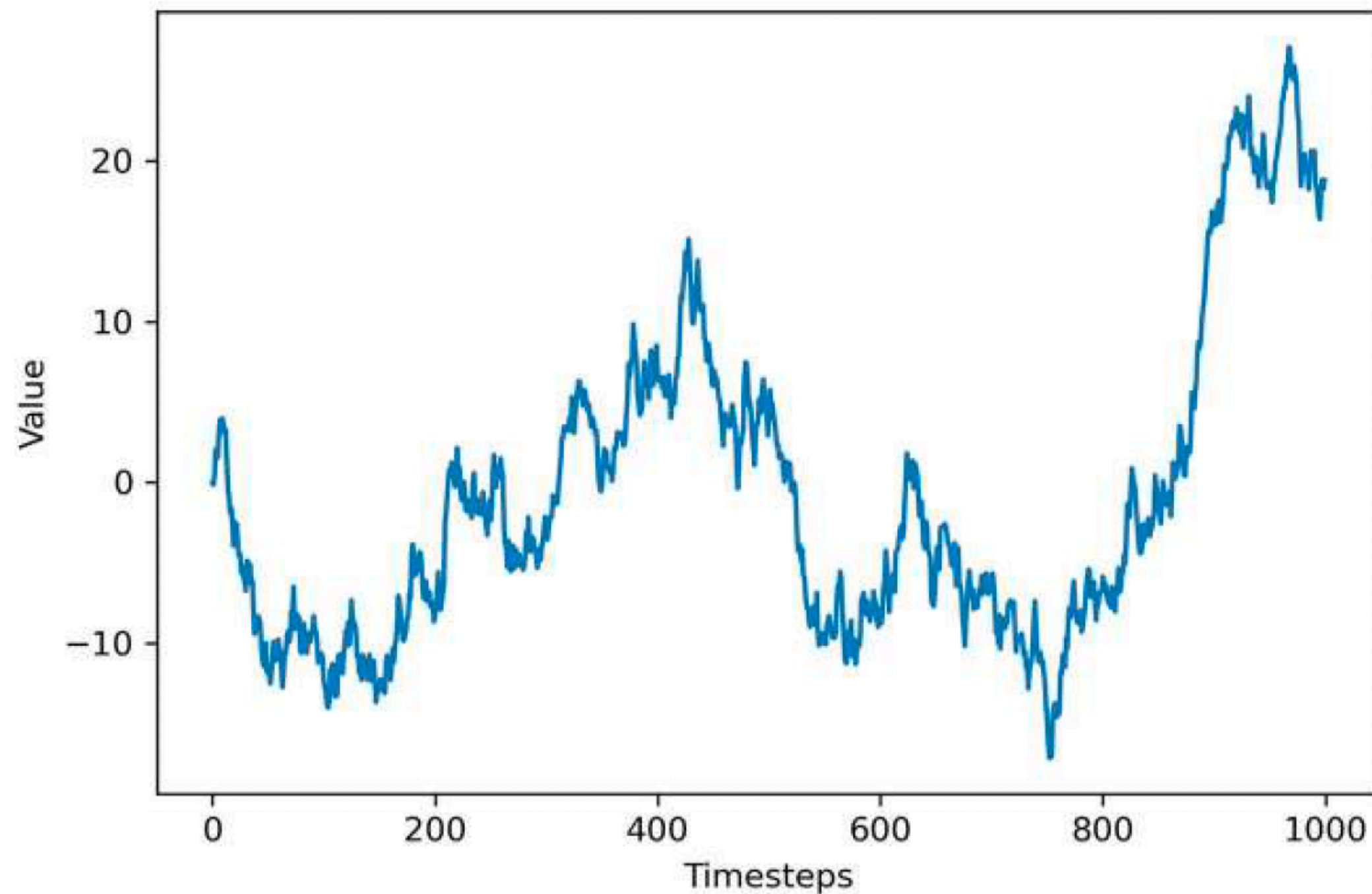
Random Walk Process

def What is a Random Walk Process?



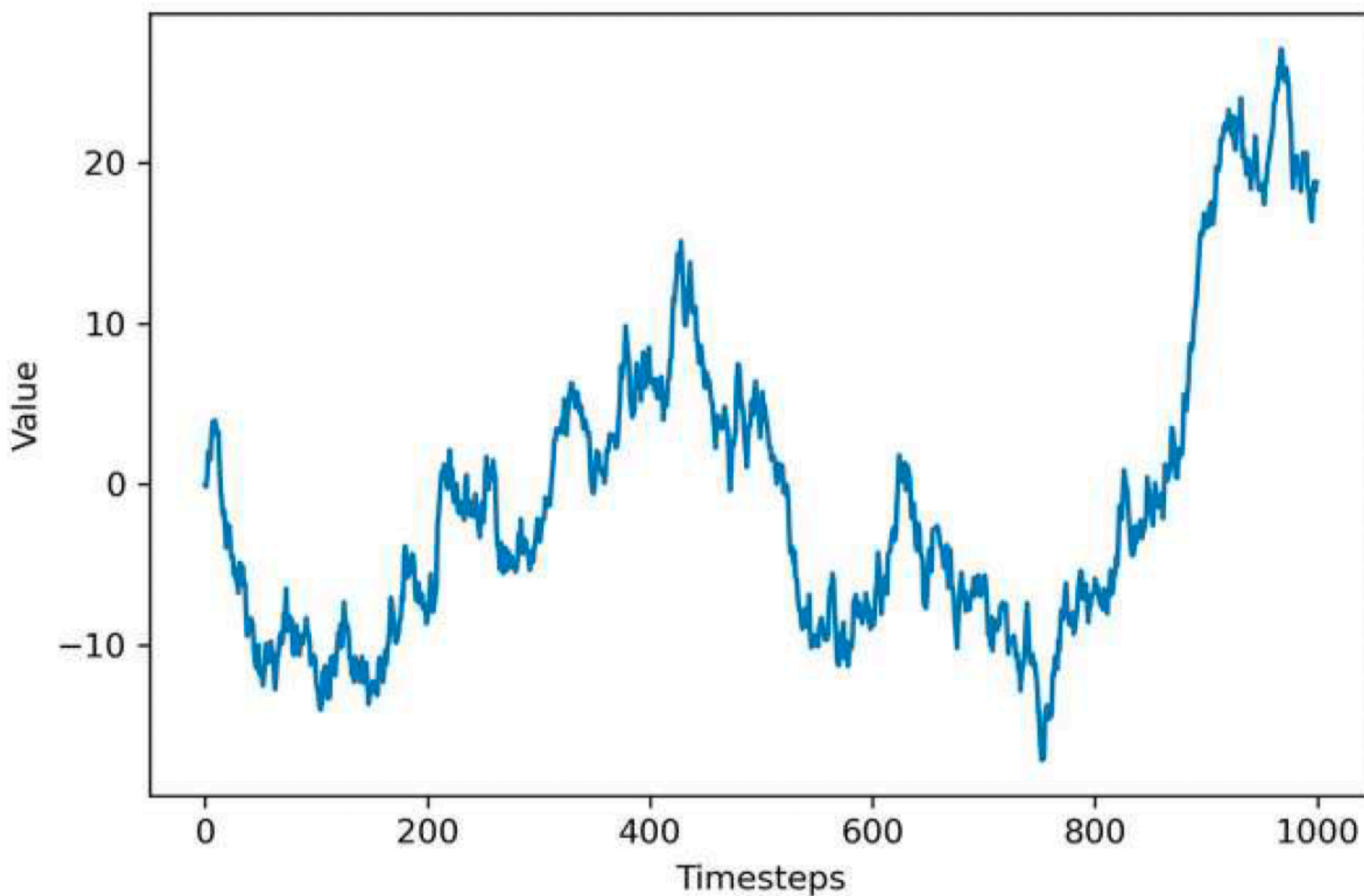
- **Random walk** is a process in which there is an equal chance of going up or down by a random number.

def What is a Random Walk Process?



- **Random walk** is a process in which there is an equal chance of going up or down by a random number.
- Usually observed in financial and economic data.
- Often exposes long periods where a positive or negative trend can be observed.
- Often accompanied by sudden changes in direction.

def What is a Random Walk Process?

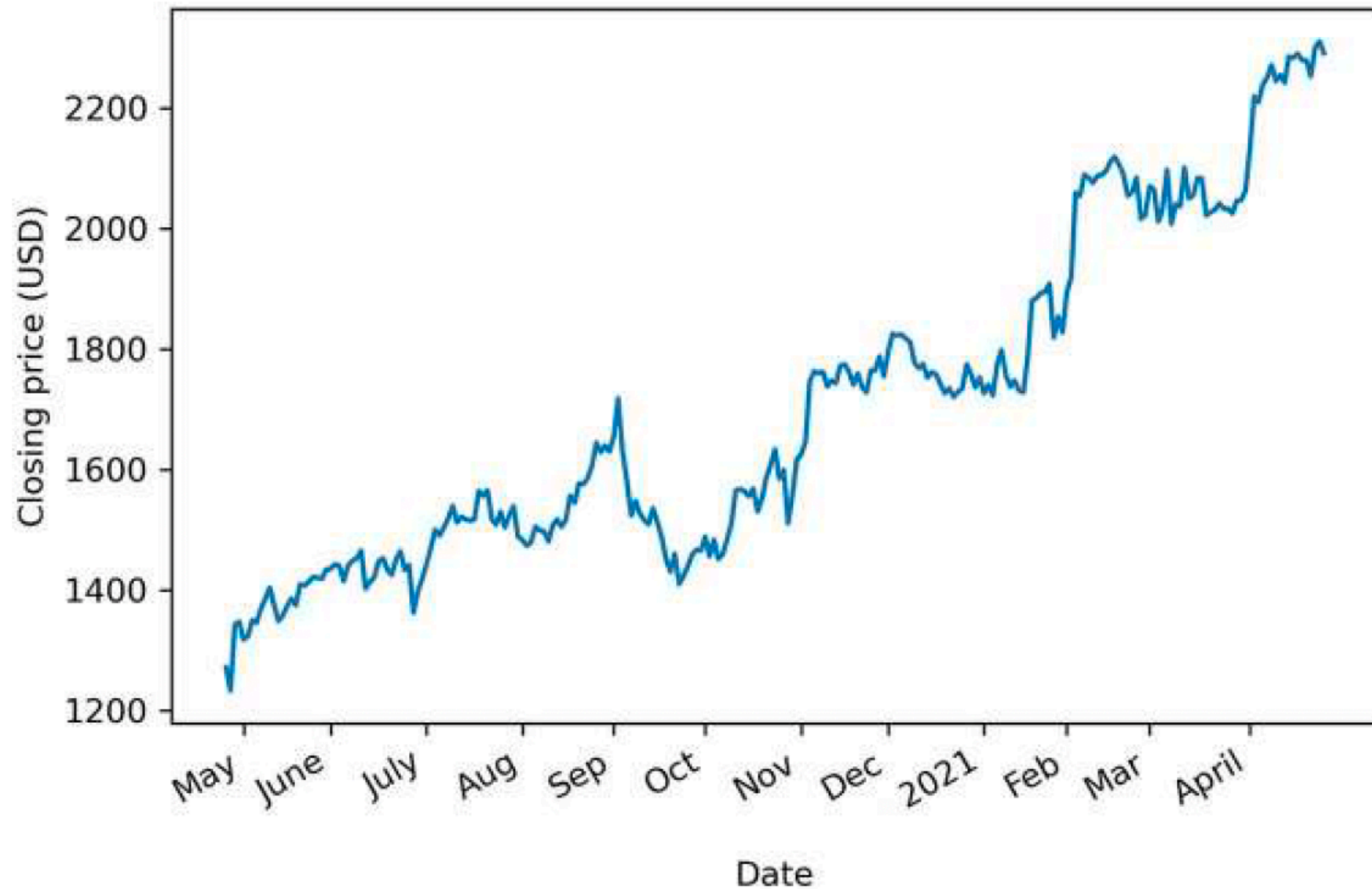


- Mathematically speaking, in a random walk process, the present value y_t is the sum of the previous timestep y_{t-1} , a constant C and a random number ϵ_t :

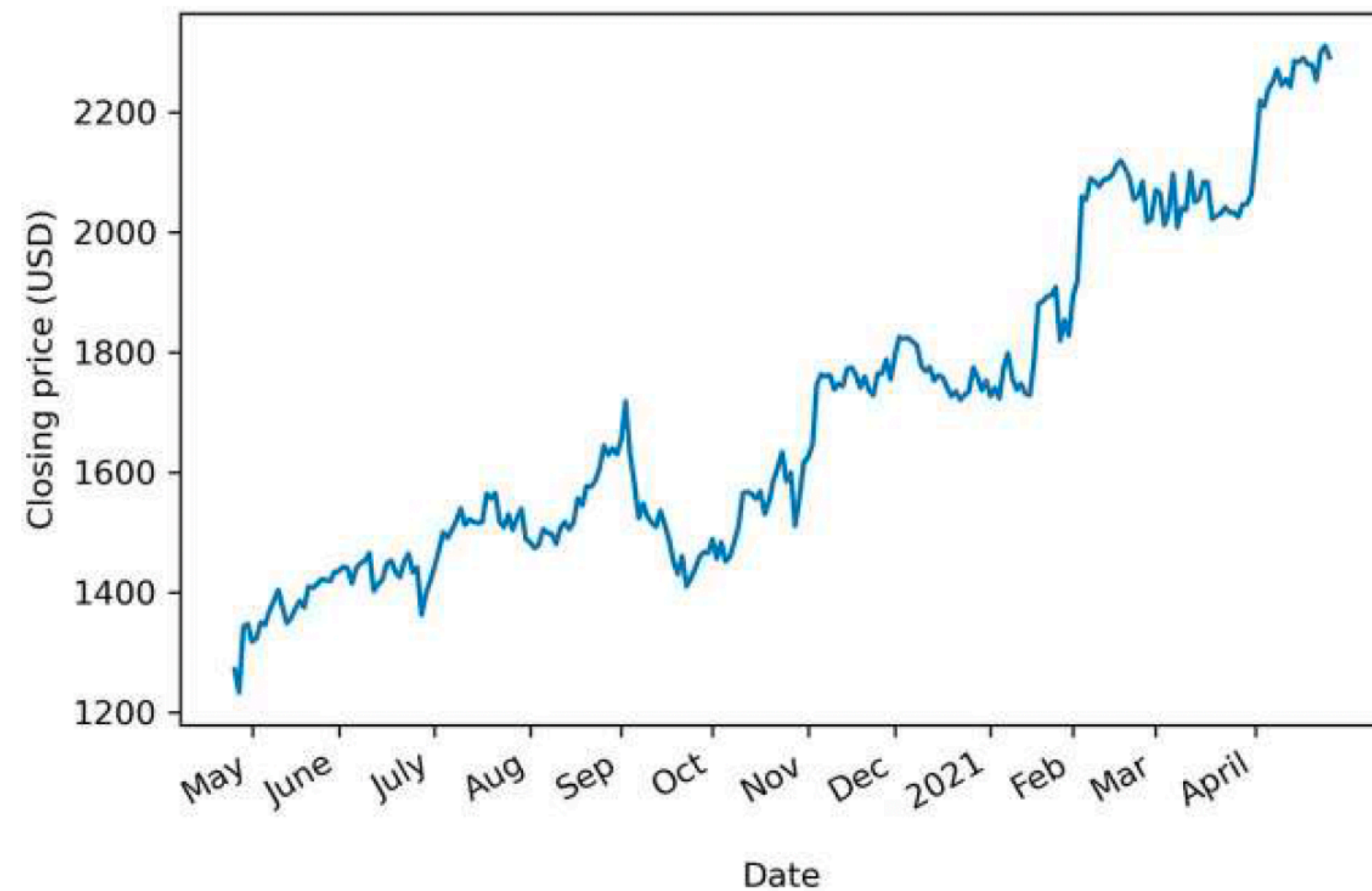
$$y_t = C + y_{t-1} + \epsilon_t.$$

The random variables $\epsilon_1, \epsilon_2, \dots$ are independent and normally distributed with mean 0 and variance 1. That is why they are also called *white noise*.

Example 1: Closing price of GOOGL from April 27, 2020, to April 27, 2021

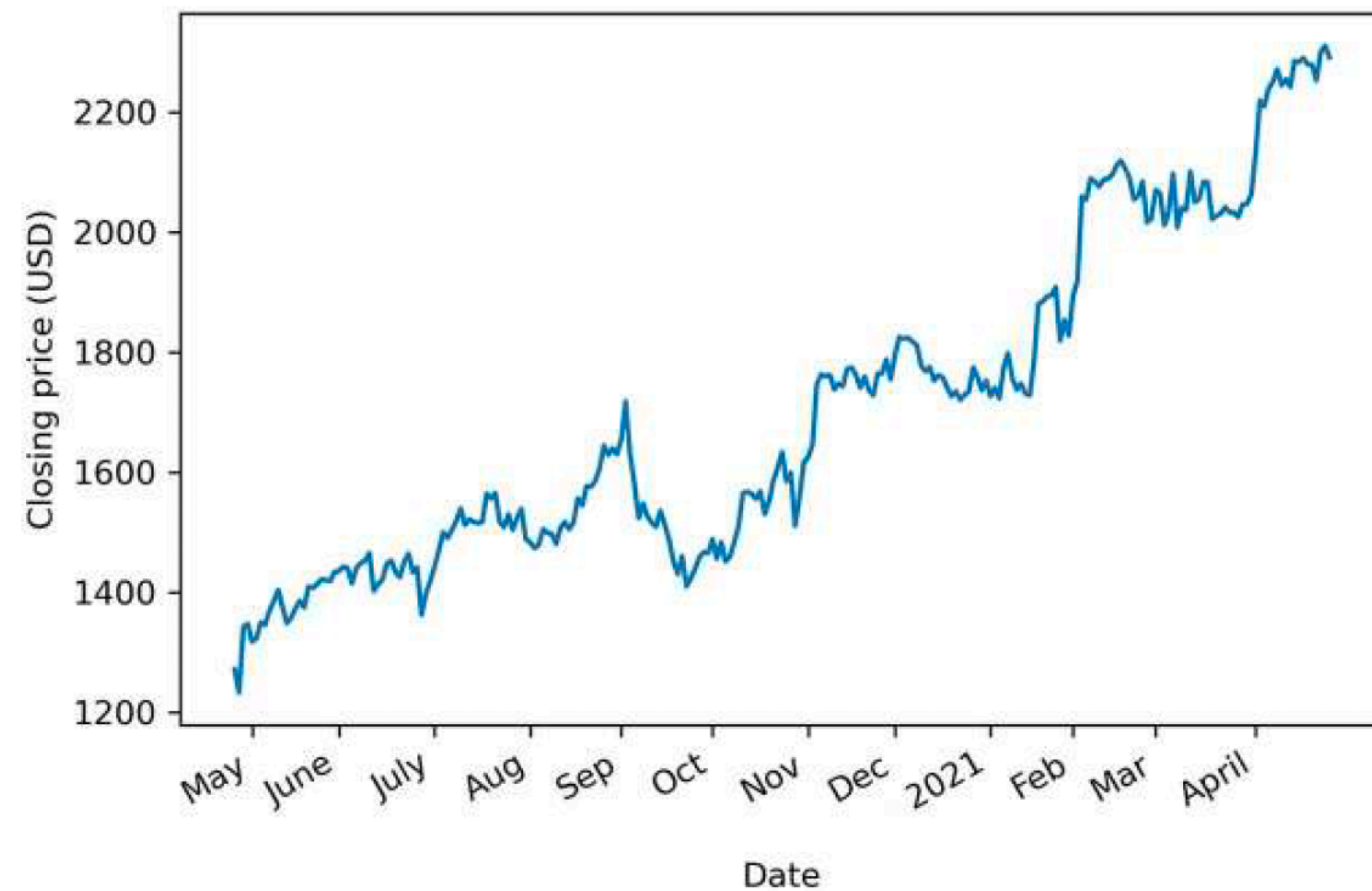


Example 1: Closing price of GOOGL from April 27, 2020, to April 27, 2021



- There is a long-term trend.
- Abrupt changes in the trend.

Example 1: Closing price of GOOGL from April 27, 2020, to April 27, 2021



- The daily closing price of GOOGL can be modelled by the random walk model.

02

Simulating a random walk

def What is a Random Walk Process?

- Mathematically speaking, in a random walk process, the present value y_t is the sum of the previous timestep y_{t-1} , a constant C and a random number ϵ_t :

$$y_t = C + y_{t-1} + \epsilon_t.$$

The random variables $\epsilon_1, \epsilon_2, \dots$ are independent and normally distributed with mean 0 and variance 1. That is why they are also called *white noise*.

- We will suppose that $C = 0$, in which case

$$y_t = y_{t-1} + \epsilon_t \quad \text{and so} \quad y_t = \epsilon_1 + \epsilon_2 + \dots + \epsilon_t.$$

Simulating a random walk

- Use the **numpy** library.

import numpy as np

Set the random seed:

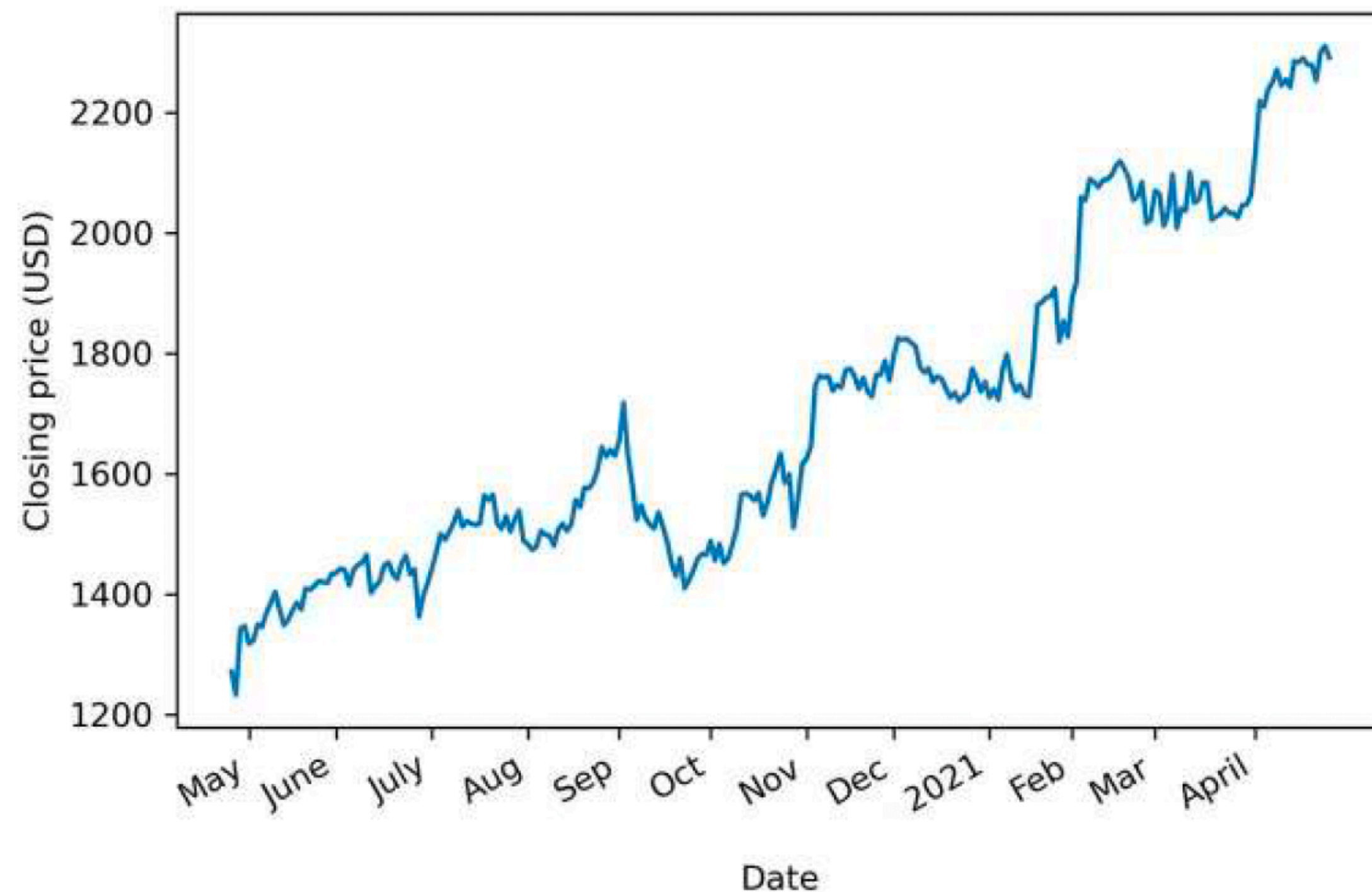
np.random.seed(42)

Generate 1000 random numbers $\sim N(0,1)$:

steps = np.random.standard_normal(1000)
steps[0]=0

The cumulative sum of errors:

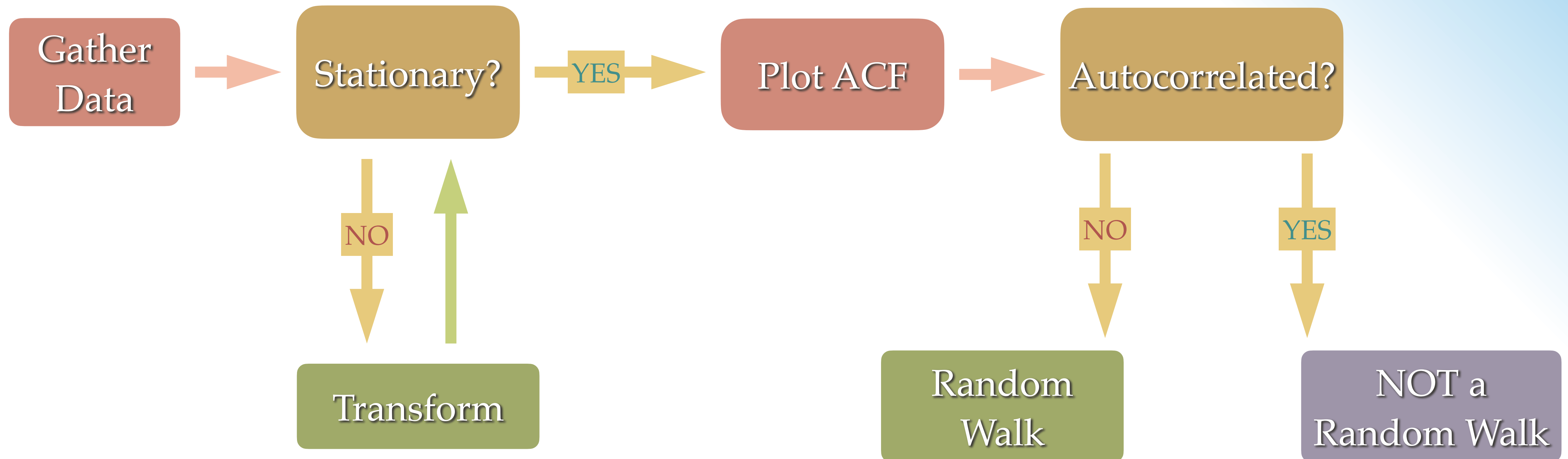
random_walk = np.cumsum(steps)



03

Identifying a random walk

Identifying a Random Walk

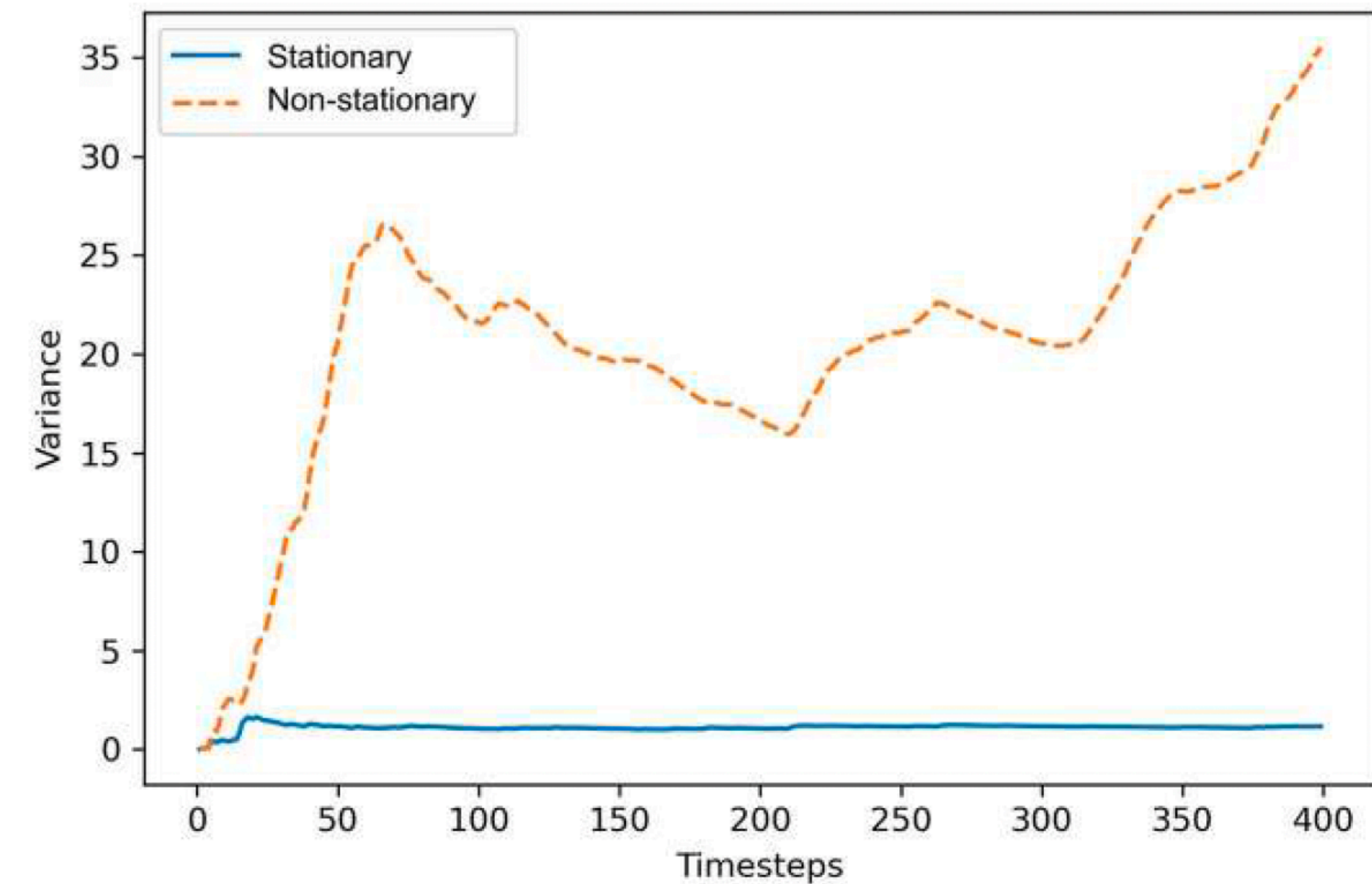


In the context of time series, **a random walk** is defined as a series whose first difference is stationary and uncorrelated.

04

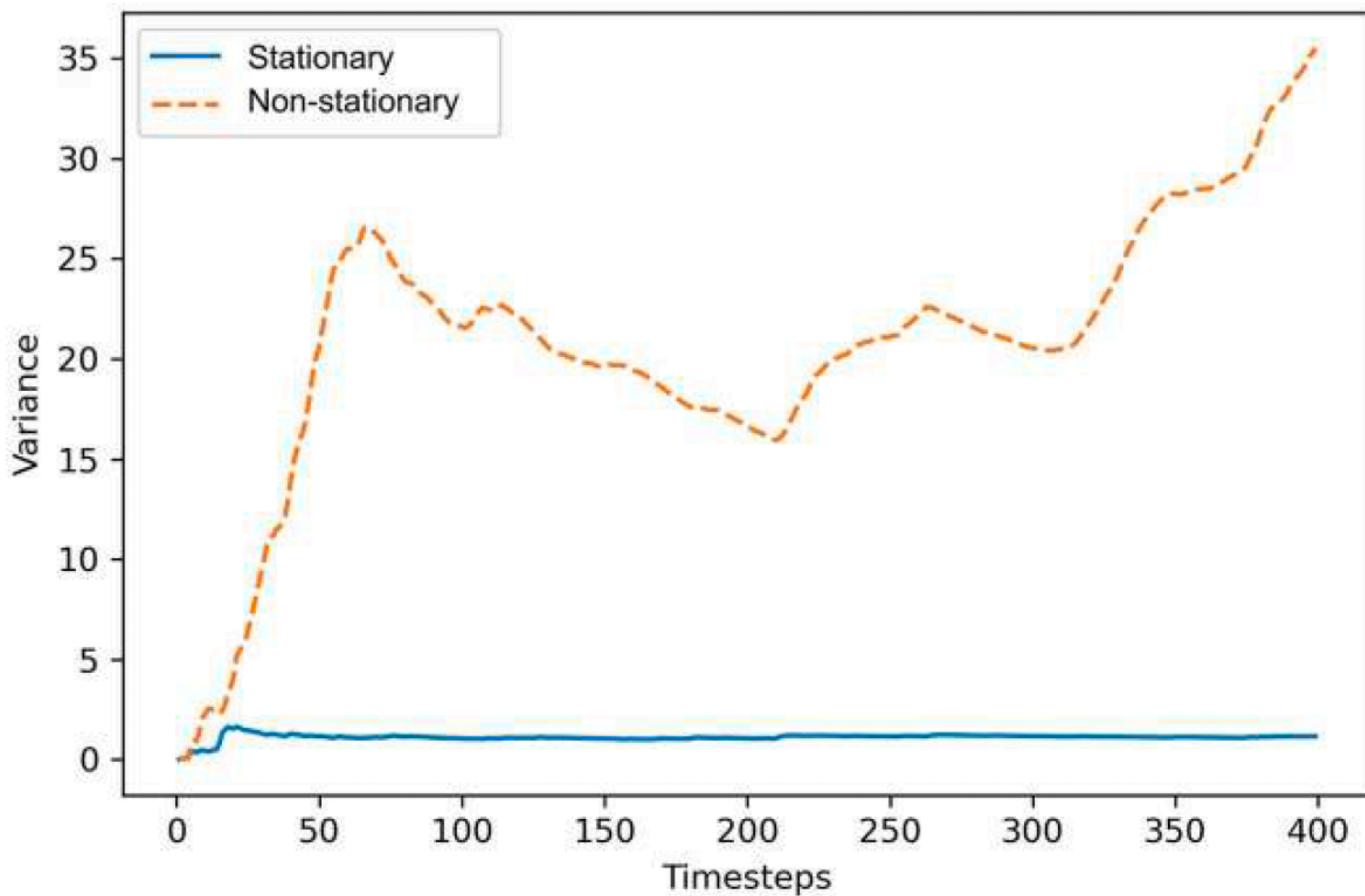
Stationarity

^{def} What is Stationarity?



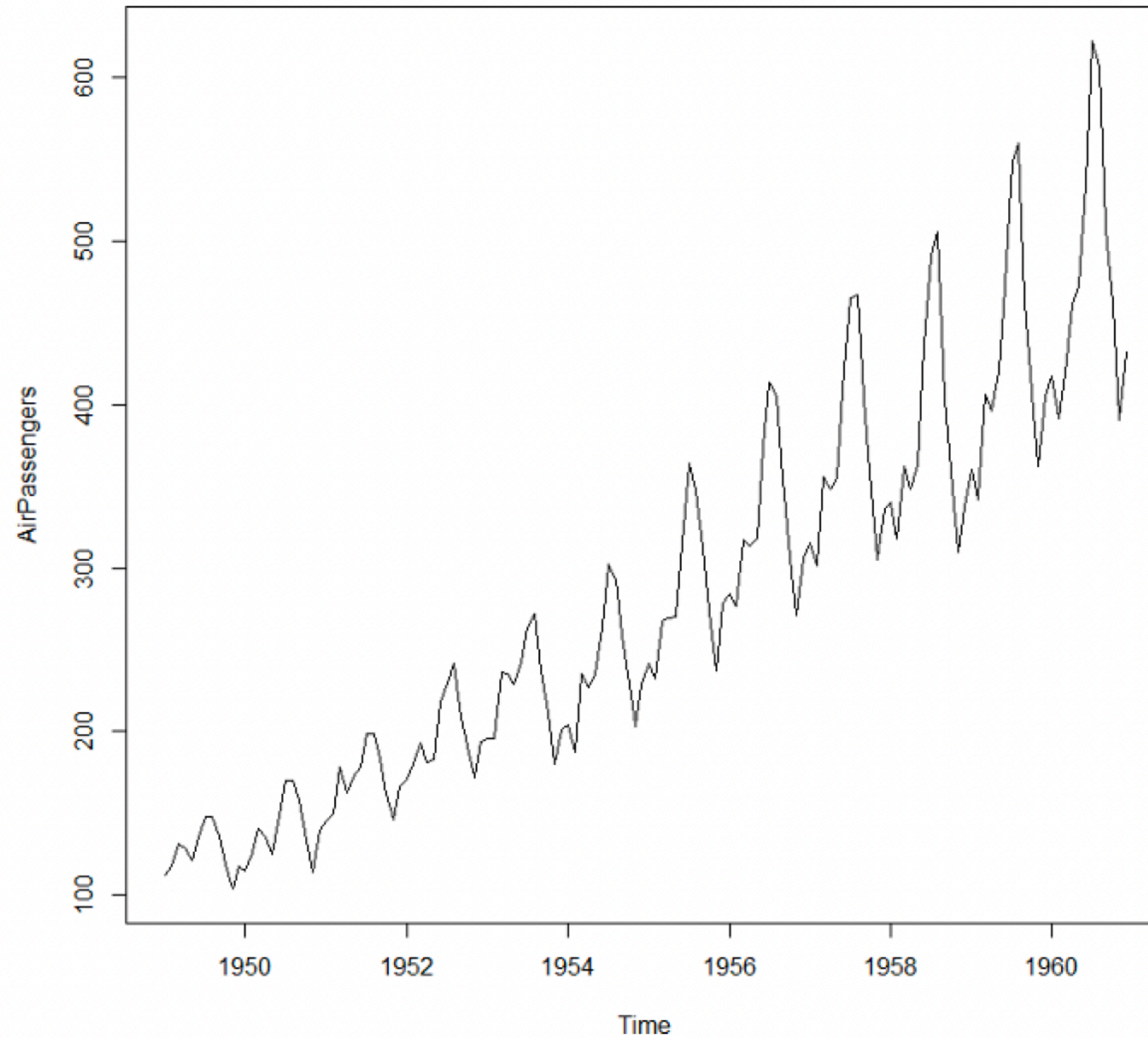
- A **stationary process** is one whose statistical properties do not change over time.
- A **stationary time series** is one that has fairly stable statistical properties over time, particularly with respect to mean and variance.

def What is Stationarity? (Intuition)

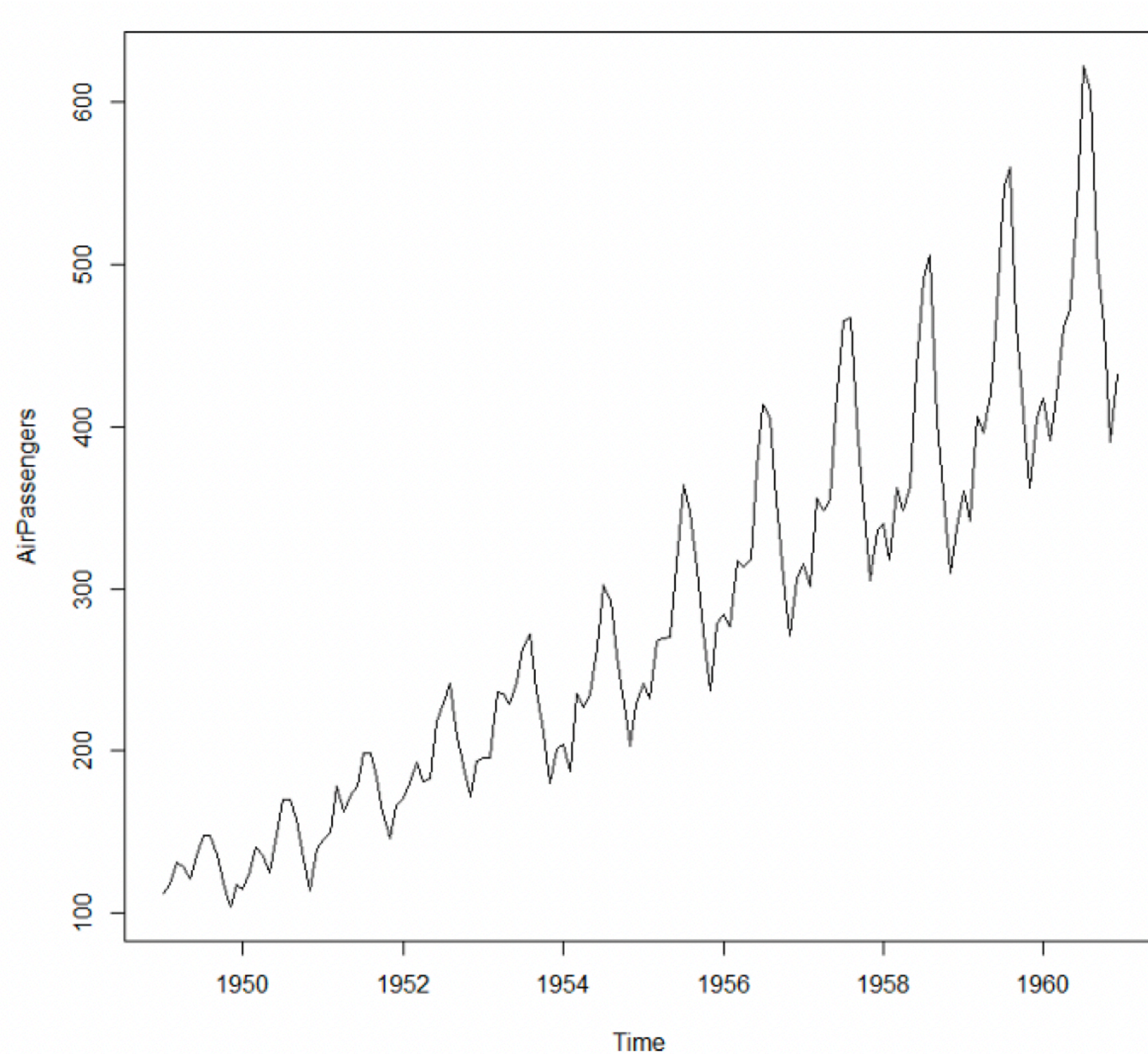


- A **stationary process** is one whose statistical properties do not change over time.
- Sometimes it can be easier to rule things out as not being stationary rather than saying something is stationary:
 1. the mean value is increasing over time, rather than remaining steady;
 2. the distance between peak and trough is growing (the variance is increasing over time);
 3. strong seasonal behaviour.

Example 2: Airline passengers data

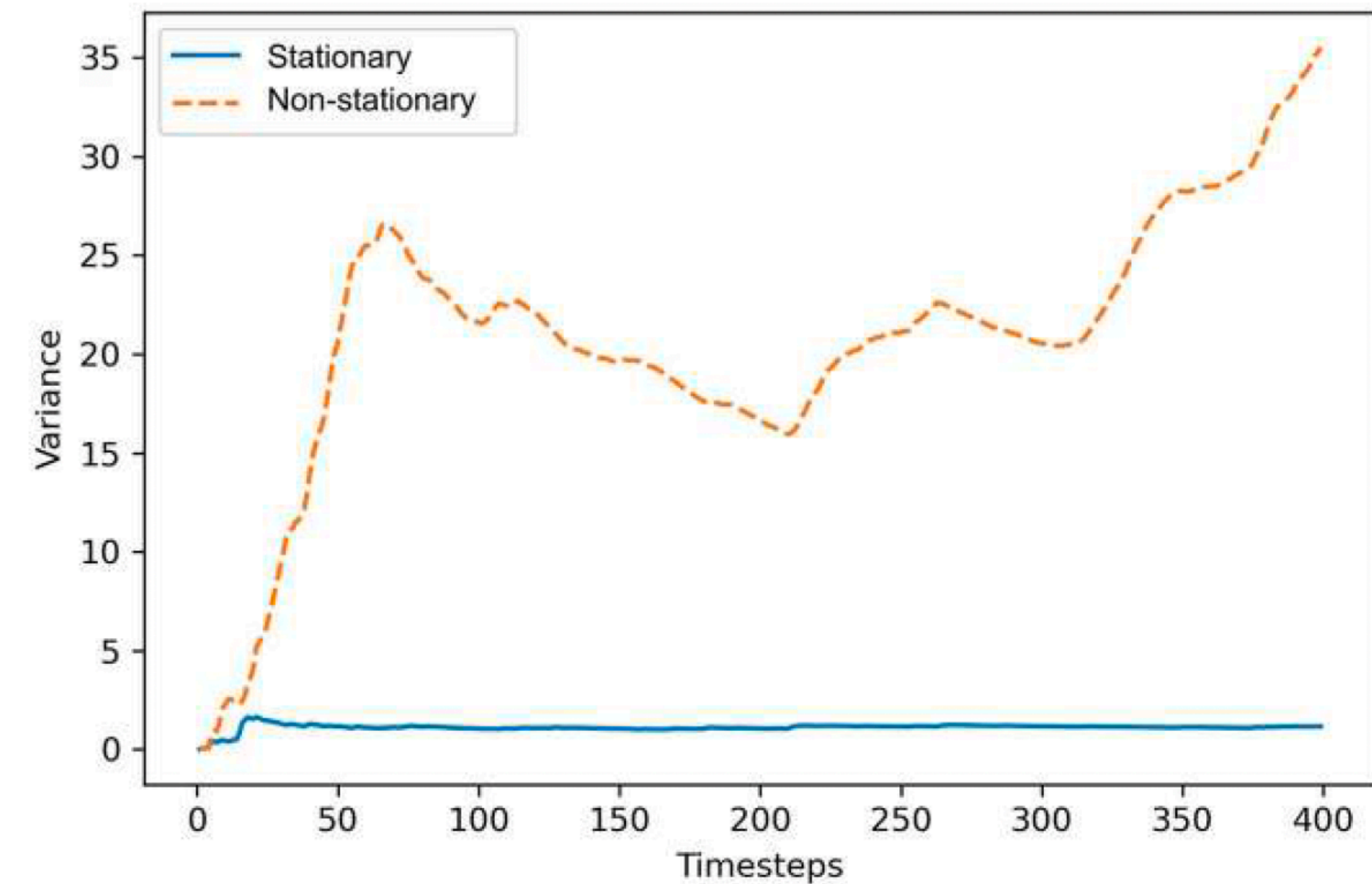


Example 2: Airline passengers data (not stationary)



- The mean is changing.
- The variance is changing.
- Evidence of seasonality.

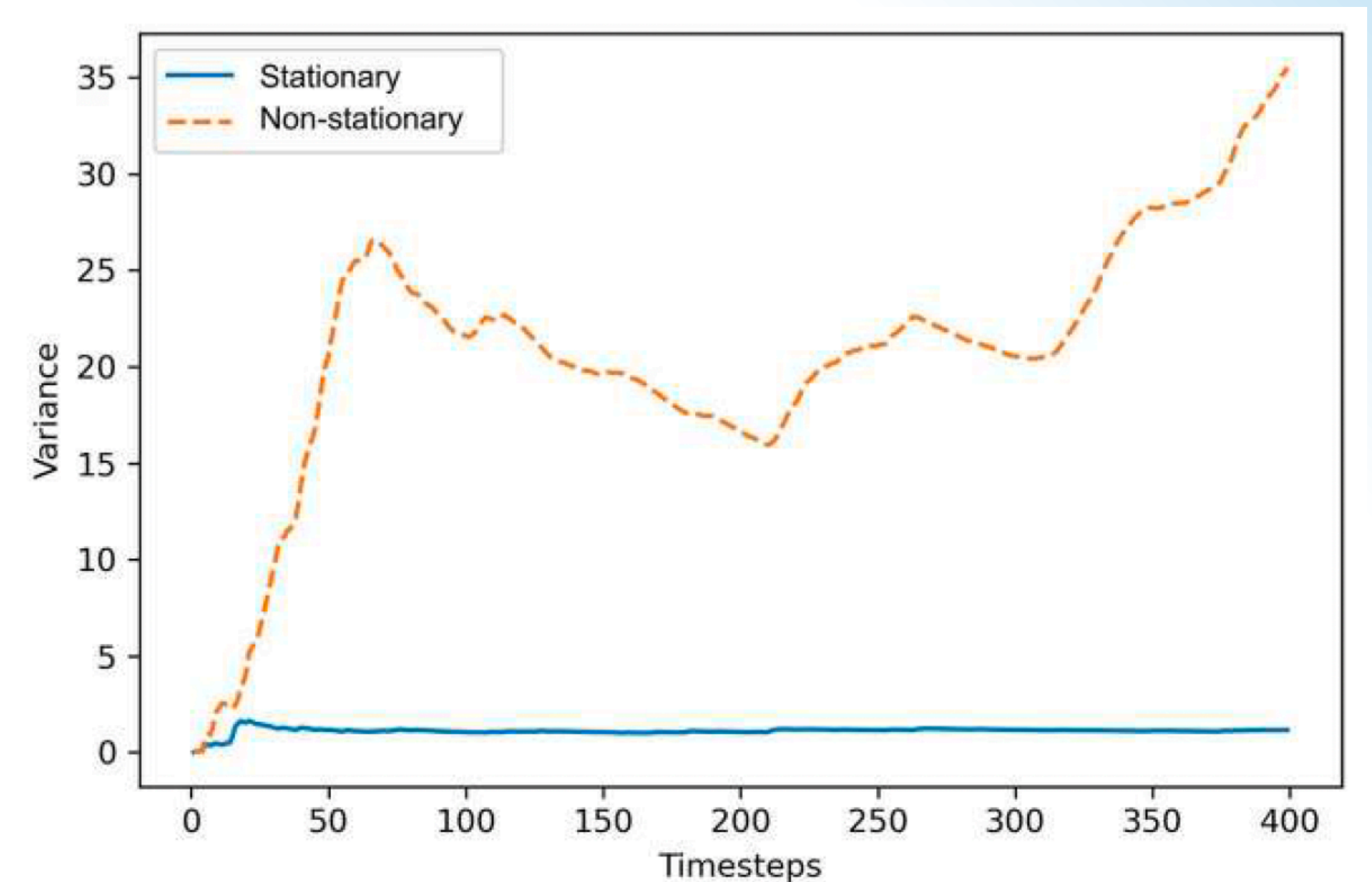
def What is Stationarity? (A formal definition)



- A process is **stationary** if for all possible lags k the distribution of $y_t, y_{t+1}, \dots, y_{t+k}$ does not depend on t .

Testing for Stationarity

- Statistical tests for stationarity often check if there is a **unit root** — that is, whether 1 is a solution of the process's characteristic equation.
- A linear time series is **non-stationary** if there is a unit root, although lack of a unit root does not prove stationarity.
- Addressing stationarity in general remains tricky, and determining if a process has a unit root remains a current area of research.

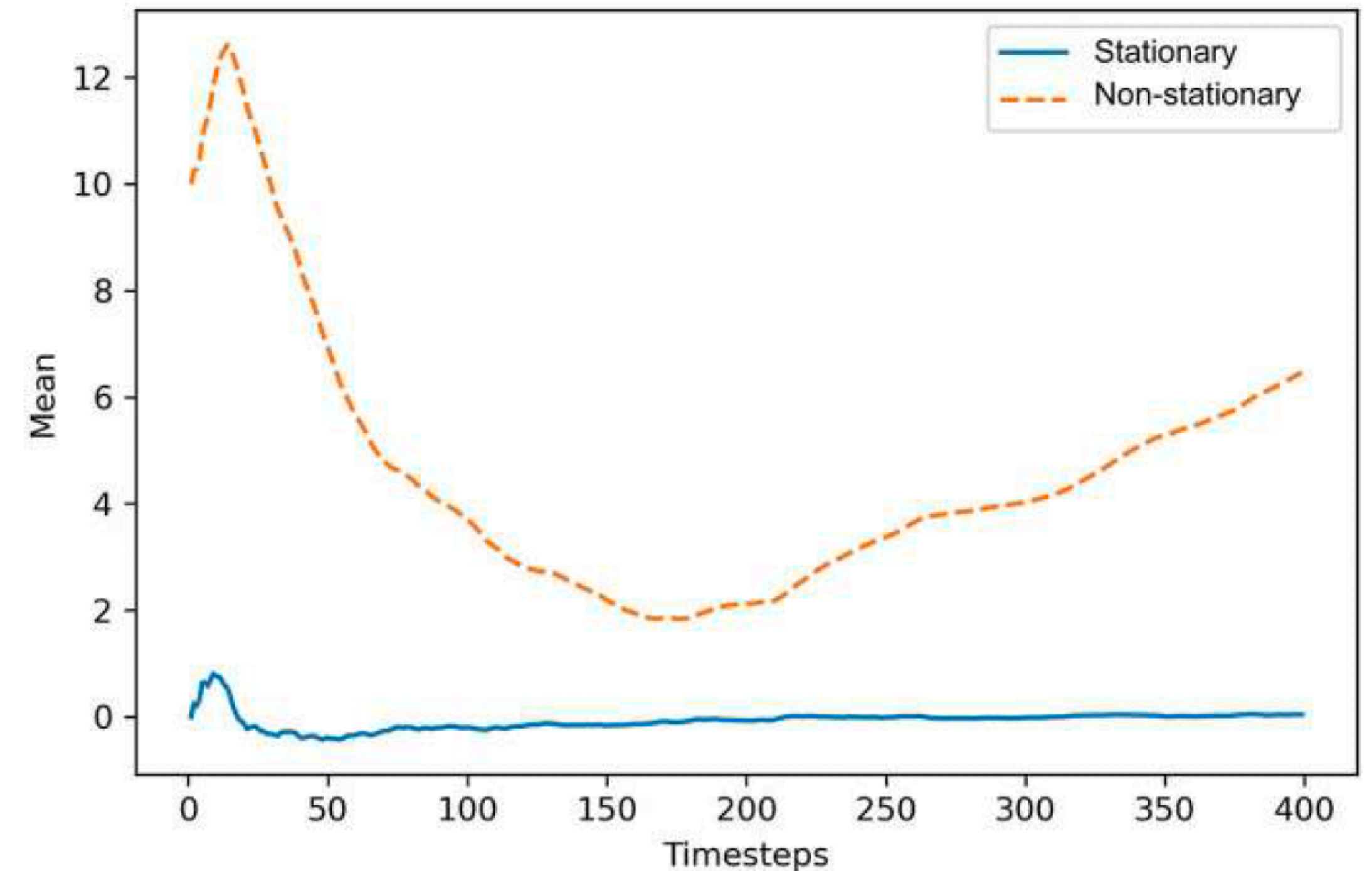


Example 3: Simulating two simple series

- The process following the equation
$$y_t = 0.5y_{t-1} + \epsilon_t$$
is stationary.
- The process following the equation
$$y_t = y_{t-1} + \epsilon_t$$
is not stationary.

For a time series with relation
$$y_t = C + \alpha y_{t-1} + \epsilon_t$$
the coefficient α is the **root** of the series:

- if $-1 < \alpha < 1$ then it is stationary;
- otherwise it is non-stationary.



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Augmented Dickey–Fuller Test

Augmented Dickey–Fuller (ADF) Test

The **augmented Dickey-Fuller (ADF)** test helps us determine if a time series is stationary by testing for the presence of a unit root.

If a unit root is present, the time series is not stationary.

- The *null hypothesis*: there is a **unit root** present in a time series.
- The *output*: **ADF statistic** (a negative number).
The more negative it is, the stronger the rejection of the null hypothesis.
- Another *output* (in Python): **p-value**.
If it is less than 0.05, then we can reject the null hypothesis (the series is stationary).

ADF Test Implementation

- Use the **statsmodels** library:

```
from statsmodels.tsa.stattools import adfuller
```

```
ADF_result = adfuller(random_walk)
```

```
print(f'ADF Statistic: {ADF_result[0]}')
```

```
print(f'p-value: {ADF_result[1]}')
```

- The *null hypothesis*: there is a **unit root** present in a time series.
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ADF Test Implementation

- Use the **statsmodels** library:

```
from statsmodels.tsa.stattools import adfuller  
ADF_result = adfuller(random_walk)  
print(f'ADF Statistic: {ADF_result[0]}')  
print(f'p-value: {ADF_result[1]}')
```

- This prints an **ADF statistic** of -0.97 and a **p-value** of 0.77 . The ADF statistic is not a large negative number, and with a p-value greater than 0.05 , we cannot reject the null hypothesis (the series is not stationary).

- The *null hypothesis*: there is a **unit root** present in a time series.
- The *output*: **ADF statistic** (a negative number).
The more negative it is, the stronger the rejection of the null hypothesis.
- Another *output* (in Python): **p-value**.
If it is less than 0.05 , then we can reject the null hypothesis (the series is stationary).

Transformation in time series forecasting

A **transformation** is a mathematical operation applied to a time series in order to make it stationary.

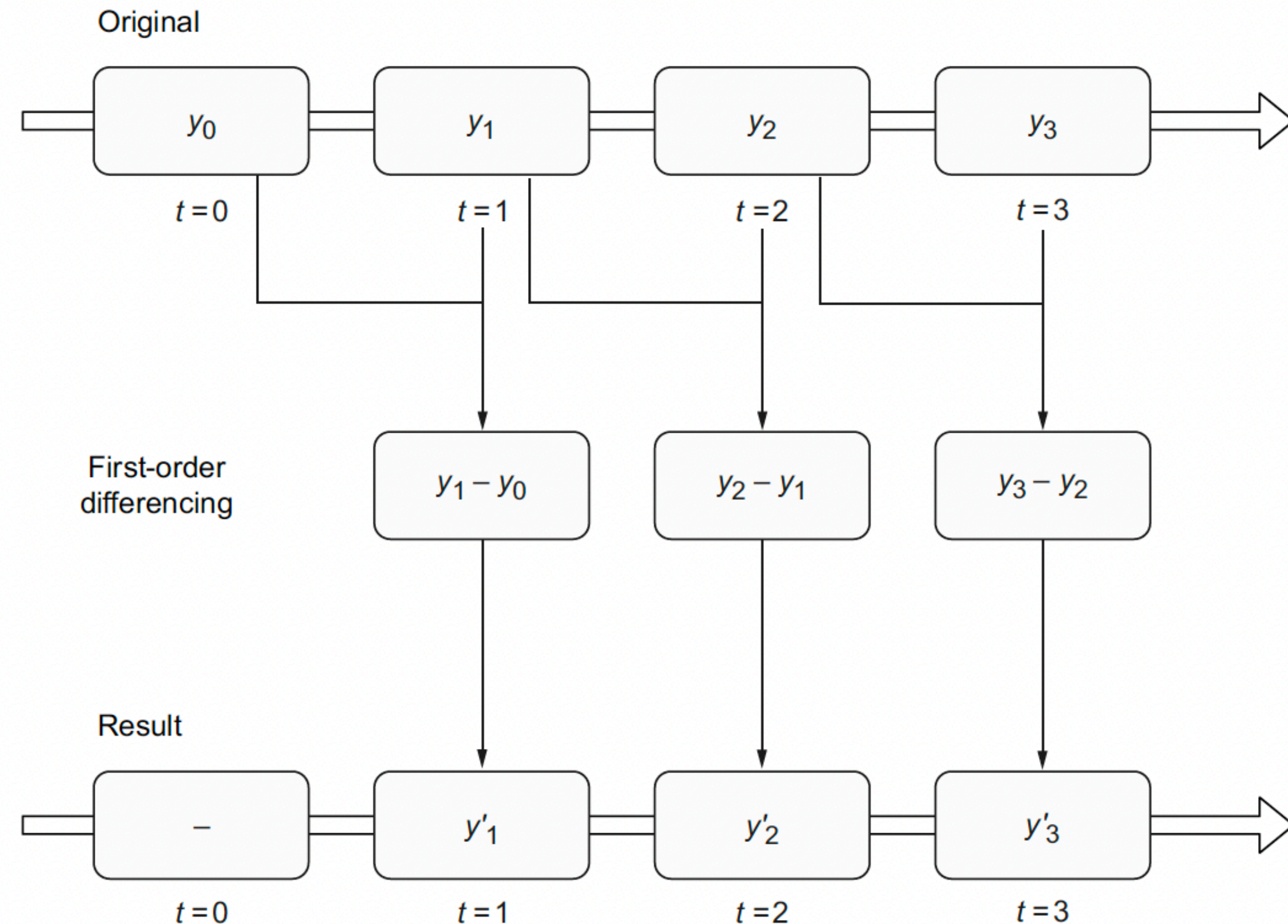
Differencing is a transformation that calculates the change from one timestep to another. It is useful for stabilising the mean.

Applying a **log** function to the series can stabilise its variance.

- In order to make a time series stationary, we can apply **differencing** (as many times as needed).
- **First-order differencing** calculates changes between neighbouring values:

$$y'_t = y_t - y_{t-1}$$

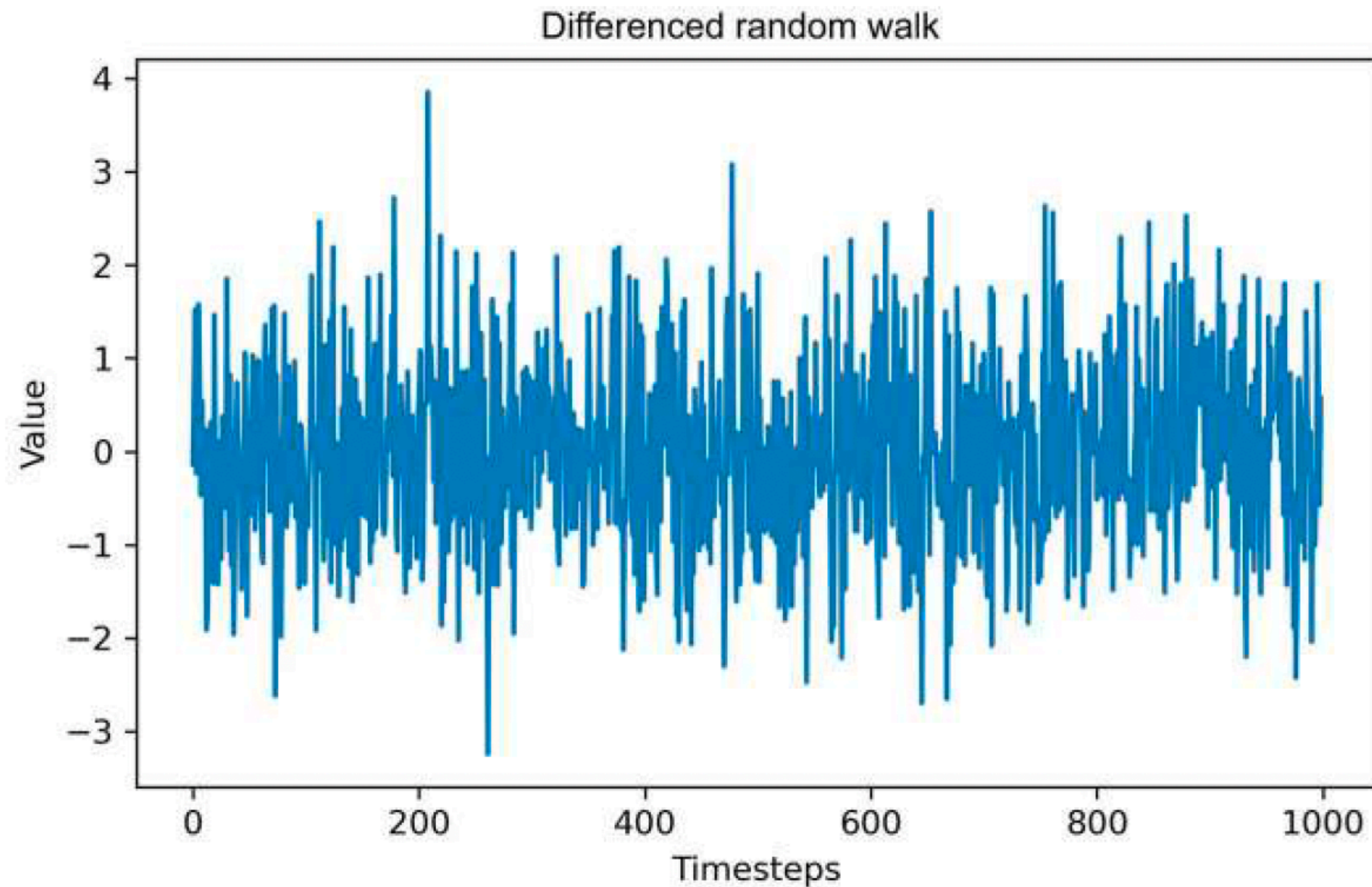
Differencing



- In order to make a time series stationary, we can apply **differencing** (as many times as needed).
- **First-order differencing** calculates changes between neighbouring values:

$$y'_t = y_t - y_{t-1}$$

Differencing (Implementation)



- Use the **numpy** library:

For the first-order differencing, the n parameter must be set to 1:

`diff_random_walk = np.diff(random_walk, n=1)`

- It seems that we successfully removed the trend and that the variance is stable.

ADF Test Implementation

- Apply the ADF test to the differenced series:

```
ADF_result = adfuller(diff_random_walk)
```

```
print(f'ADF Statistic: {ADF_result[0]}')
```

```
print(f'p-value: {ADF_result[1]}')
```

- The *null hypothesis*: there is a **unit root** present in a time series.
- The *output*: **ADF statistic** (a negative number).
The more negative it is, the stronger the rejection of the null hypothesis.
- Another *output* (in Python): **p-value**.
If it is less than 0.05, then we can reject the null hypothesis (the series is stationary).

ADF Test Implementation

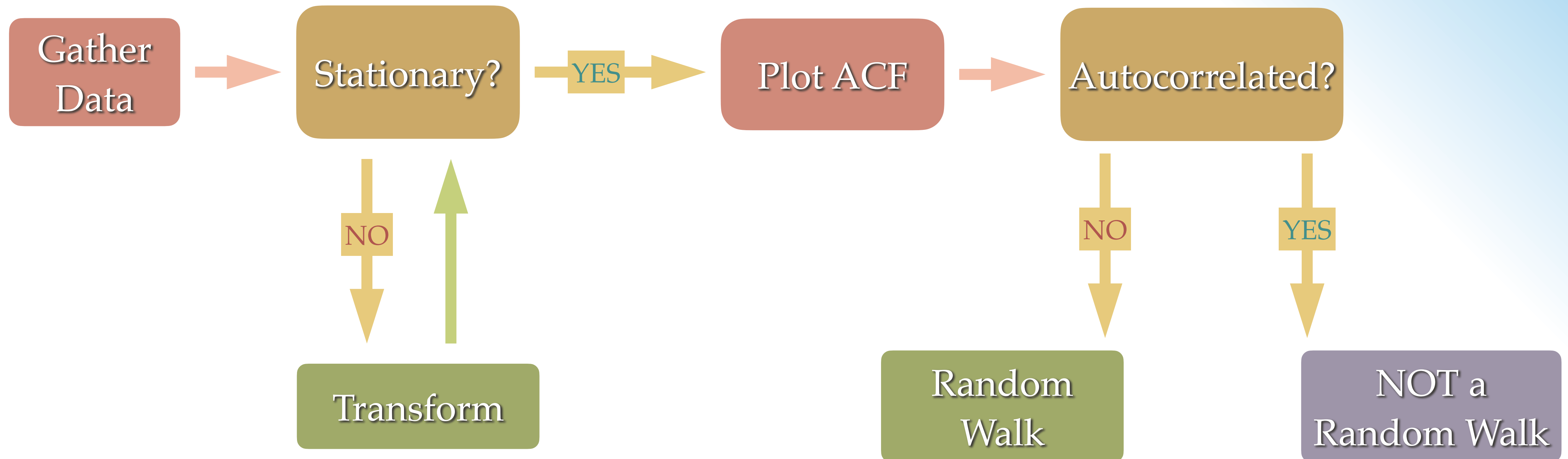
- Apply the ADF test to the differenced series:

```
ADF_result = adfuller(diff_random_walk)  
print(f'ADF Statistic: {ADF_result[0]}')  
print(f'p-value: {ADF_result[1]}')
```

- This prints out an **ADF statistic** of -31.79 with a **p-value** of 0. This time the ADF statistic is a large negative number, and the p-value is less than 0.05.
- Therefore, we reject the null hypothesis, and we can say that this process has no unit root and is thus *stationary*.

- The *null hypothesis*: there is a **unit root** present in a time series.
- The *output*: **ADF statistic** (a negative number).
The more negative it is, the stronger the rejection of the null hypothesis.
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Identifying a Random Walk

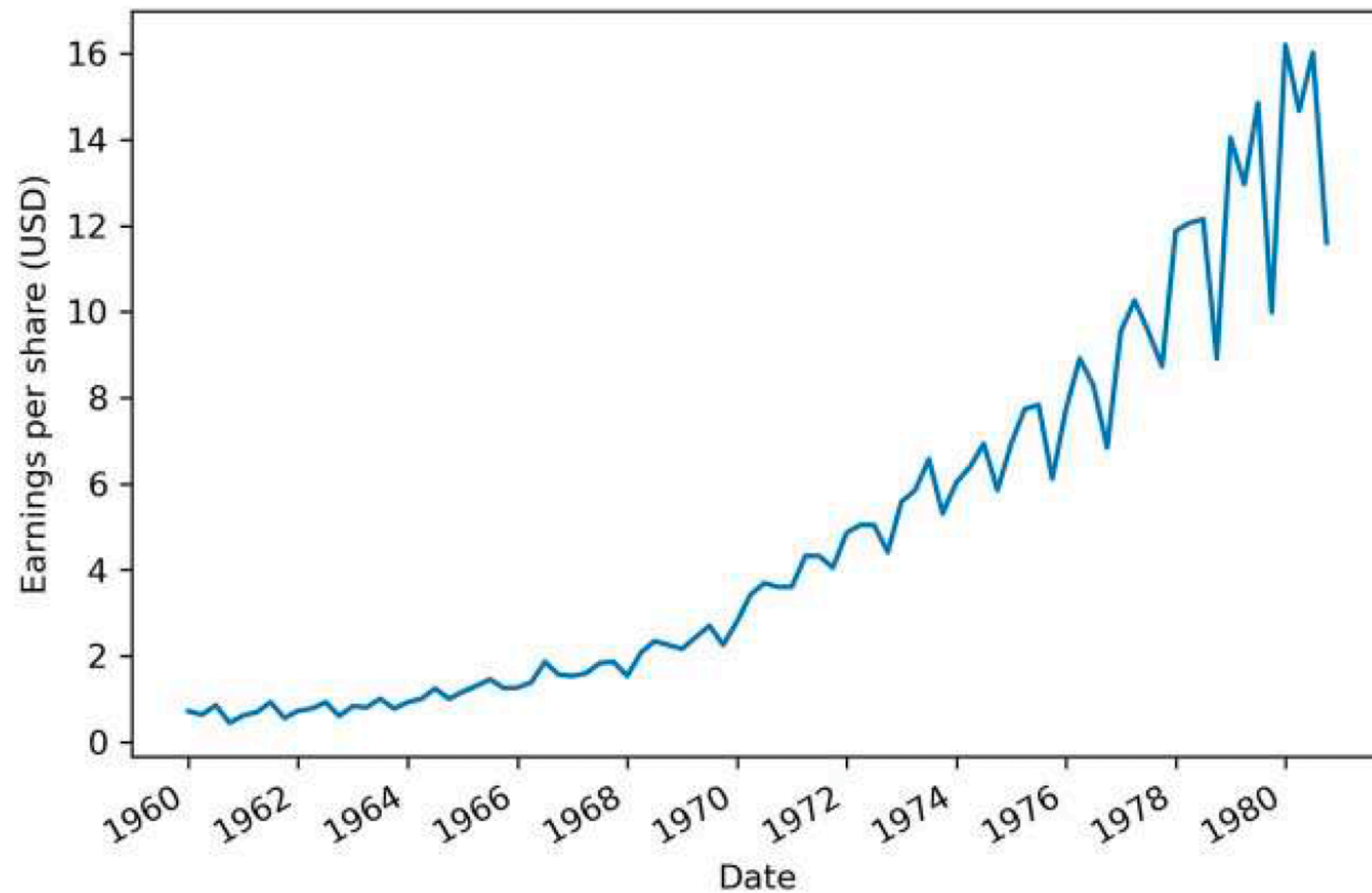


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06

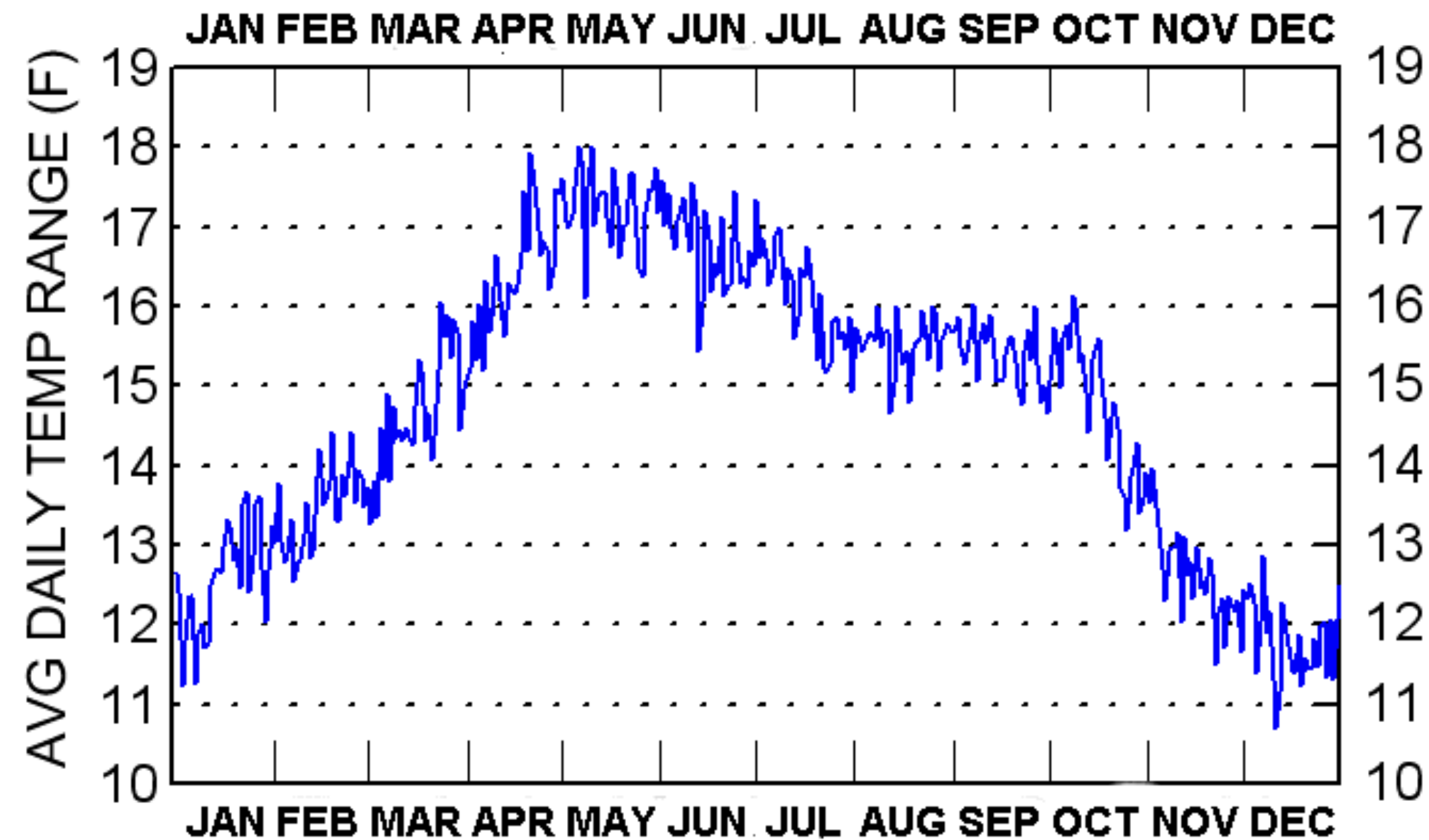
Autocorrelation

^{def} What is Autocorrelation?



- **Autocorrelation** of a time series is the idea that a value in a time series at one given point in time may have a correlation to the value at another point in time. It is the similarity between observations as a function of the time lag between them.
- The **lag** is the number of timesteps separating two values.
- More formally, **autocorrelation** is the correlation of a value with a delayed copy of itself as a function of the delay.

Example 4: Daily temperature

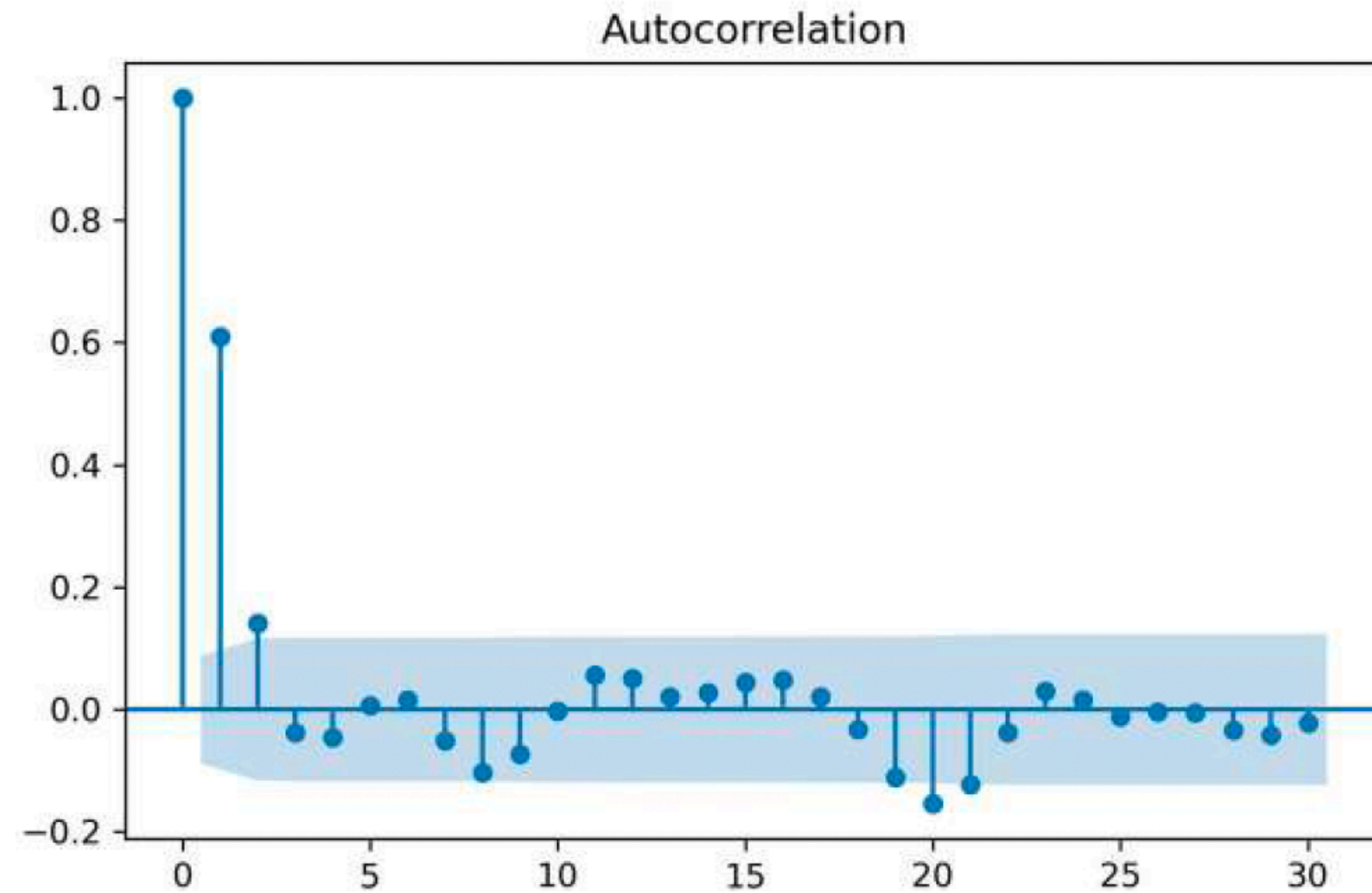


- Take a yearly time series of daily temperature data. You may find that comparing May 24th of every year to June 24th of every year will give you some correlation, such that hotter May 24ths tend to correlate with hotter June 24ths (or tend to correlate with cooler June 24ths).

07

Autocorrelation Function (ACF)

^{def} Autocorrelation Function (ACF)



- The **autocorrelation function (ACF)** measures the linear relationship between lagged values of a time series.
- For example, we can calculate the autocorrelation between y_t and y_{t-2} . Then the lag would be 2 and the autocorrelation coefficient would be some number.

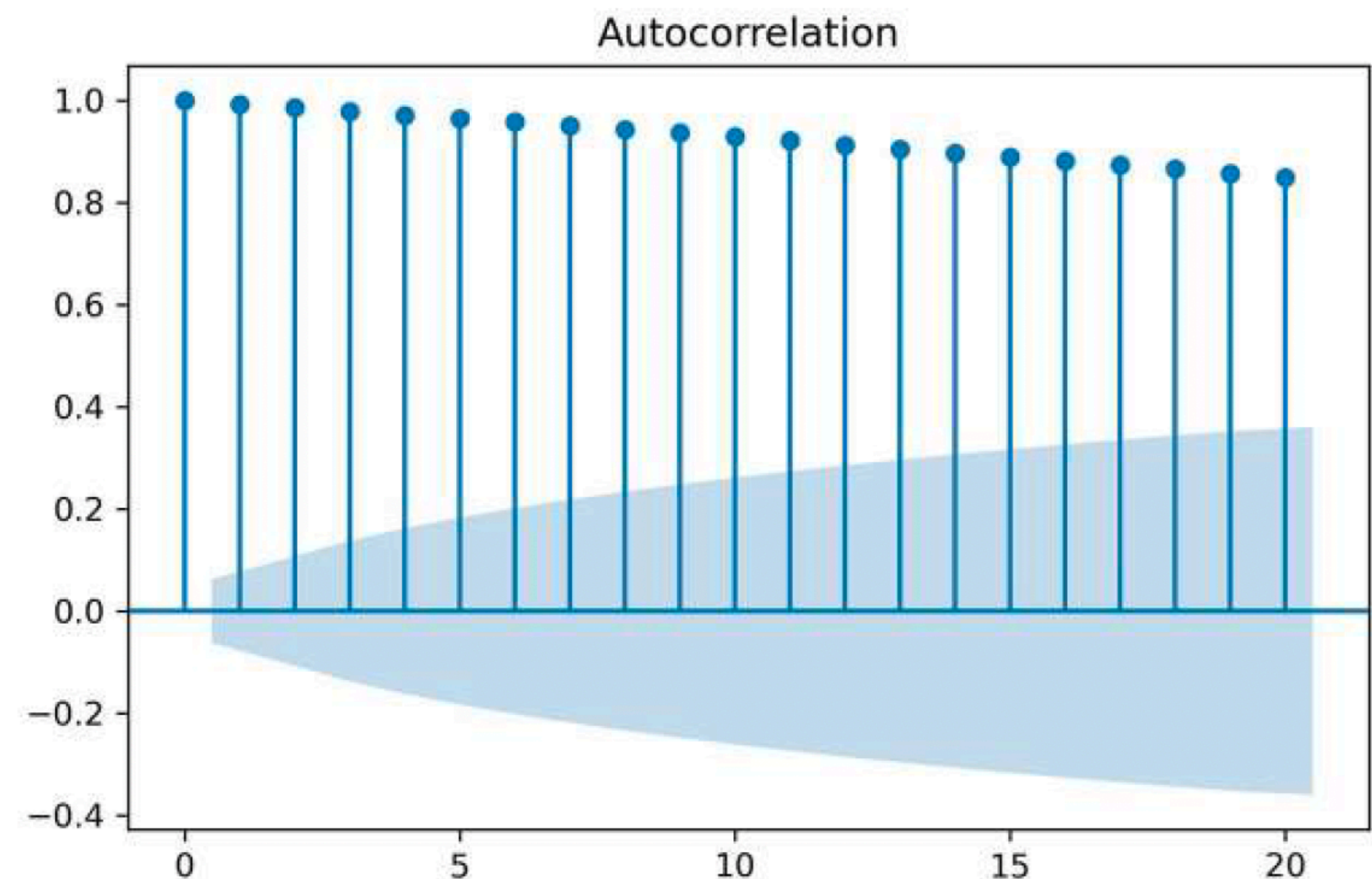
ACF Implementation

- Use the **statsmodels** library:

from statsmodels.graphics.tsaplots import plot_acf

***# Optionally specify the number of lags, which will
determine the range on the x-axis:***

plot_acf(random_walk, lags=20)

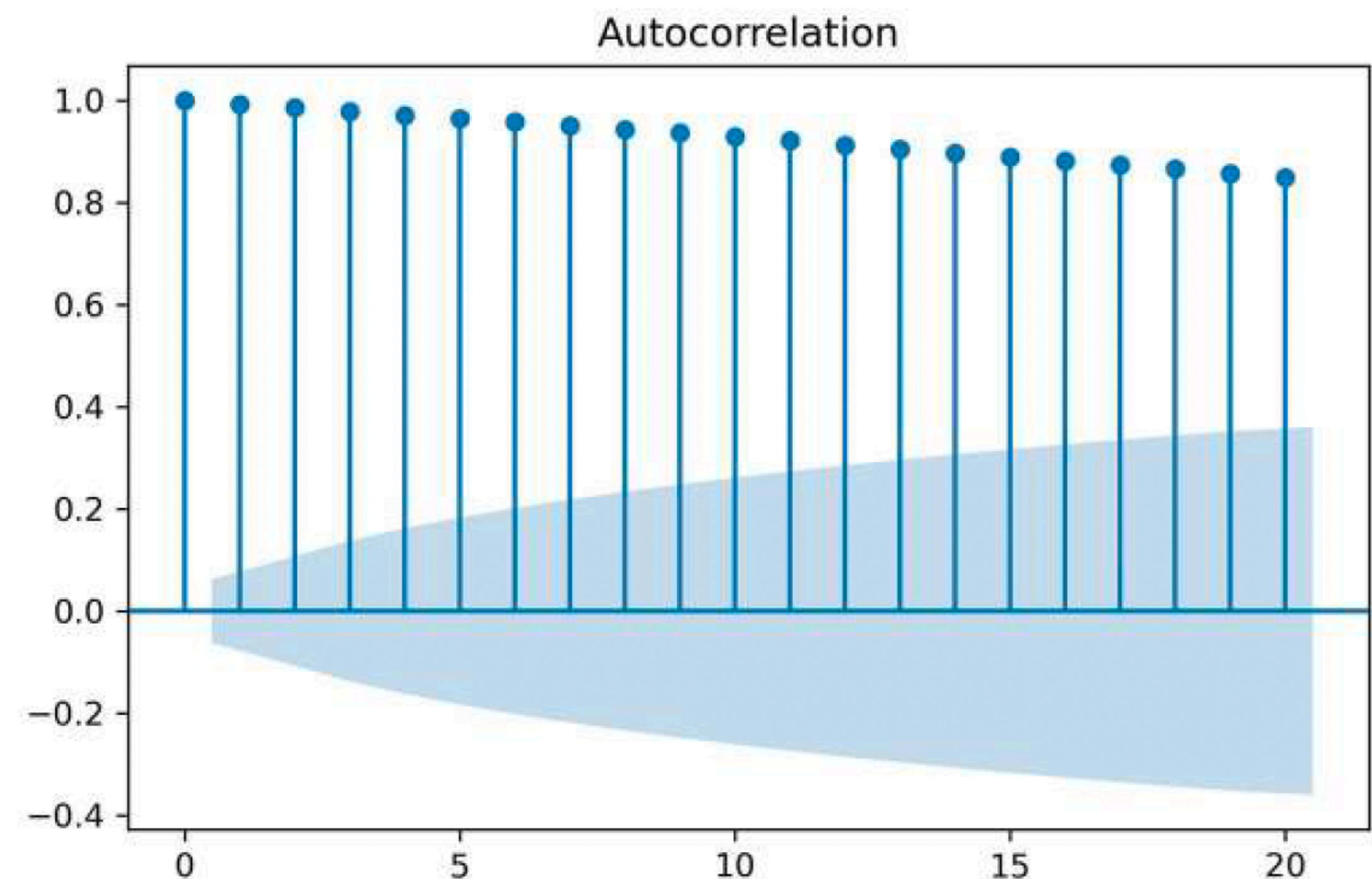


ACF Implementation

- Use the **statsmodels** library:

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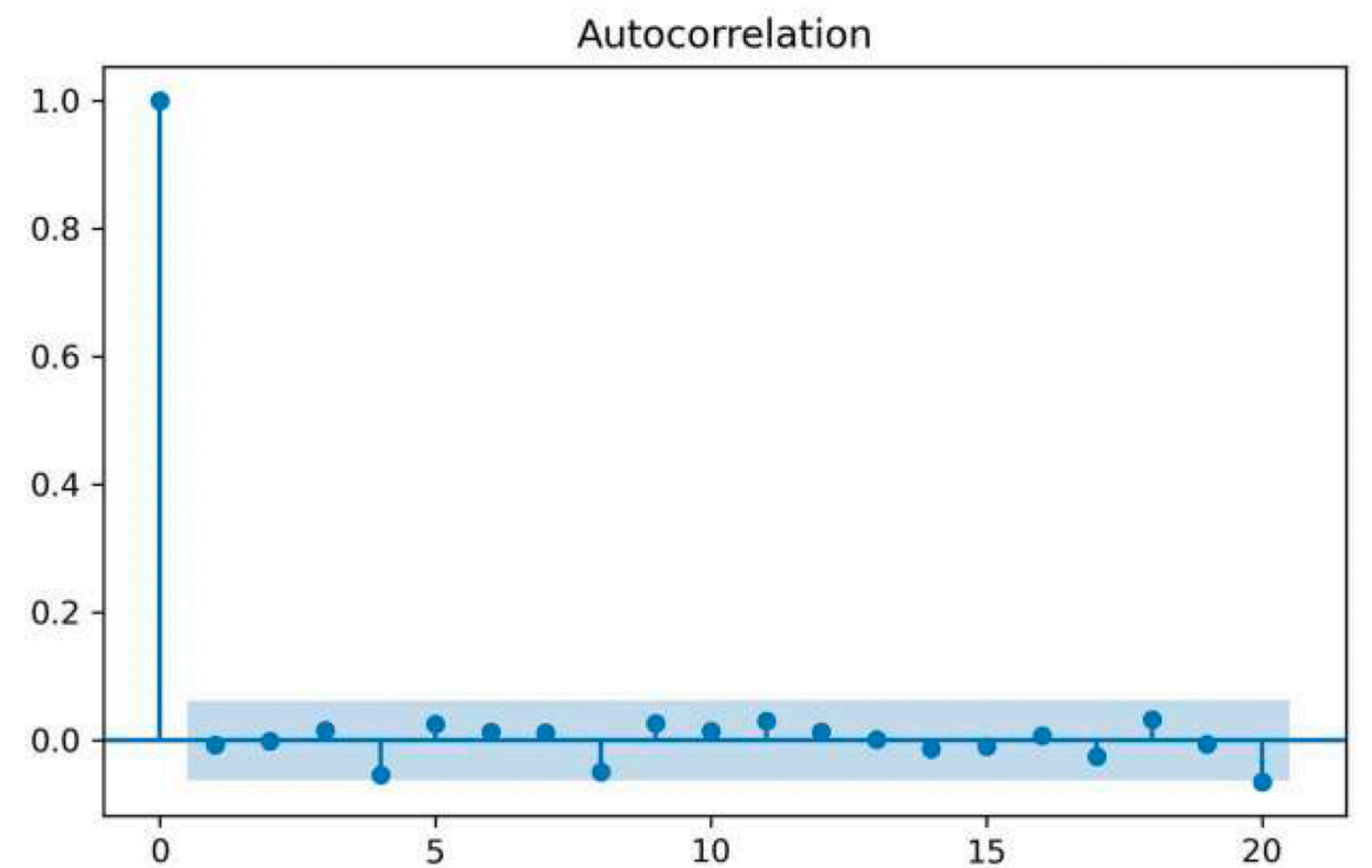
- The autocorrelation coefficients slowly decrease as the lag increases, which is a clear indicator that our random walk is **not** a stationary process.
- The shaded area represents a confidence interval. If a point is within the shaded area, then it is not significantly different from 0. Otherwise, the autocorrelation coefficient is significant.



ACF Implementation (After differencing)

- Plot the ACF for the differenced series:

`plot_acf(diff_random_walk, lags=20)`

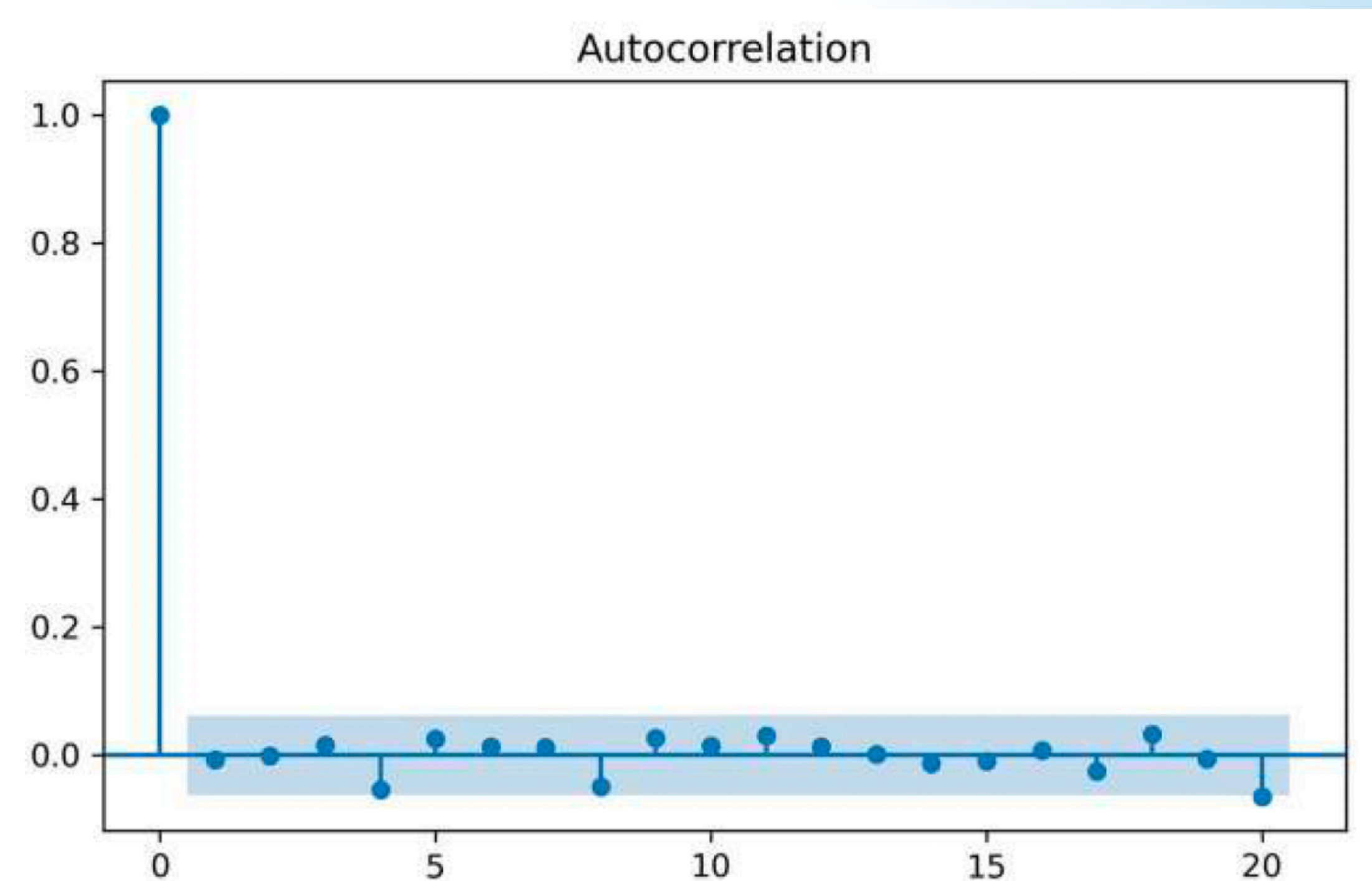


ACF Implementation (After differencing)

- Plot the ACF for the differenced series:

plot_acf(diff_random_walk, lags=20)

- There are no significant autocorrelation coefficients after lag 0.
- This means that the stationary process is completely random and can therefore be described as white noise. Each value is simply a random step away from the previous one, with no relation between them.



What is next?

- We will apply baseline models to make forecasts for a random walk.
- What happens if your process is stationary and autocorrelated, meaning that you see consecutive significant coefficients on the ACF plot? In such a situation, we will be facing a process that can be approximated by the **moving average (MA) model**, an **autoregressive (AR) model**, or the combination of both processes, leading to an **autoregressive moving average (ARMA) model**.

Thank you!